

MINI GACS PROTOTYPE — FINAL REPORT

GenTA Competency Assessment — AI R&D Engineer

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Date: 27 February 2026

Executive Summary

This project implements a compact, reproducible prototype of a GACS-like affective computing pipeline to estimate the perceived “vibe” (mood/style) of short art and marketing video clips. The system extracts representative frames, generates CLIP-based visual embeddings, computes cosine similarity, and provides retrieval and visualization tools (heatmap, top-k neighbors, and 2D projection).

The design follows a verification-first methodology with structured testing, embedding sanity checks, and documented AI usage governance. The prototype serves as a defensible foundation for a production-grade GACS engine capable of linking creative style to performance outcomes.

1. Problem Framing

GenTA’s objective is to make contemporary art and marketing visuals emotionally interpretable and to connect creative qualities to measurable performance signals. A required capability is automated “vibe” estimation for images and short clips to:

- Group creatives by stylistic similarity
- Retrieve mood-consistent variants for A/B testing
- Enable future linkage between embeddings and metrics such as CTR/ROAS

This prototype demonstrates a minimal yet defensible solution: use pre-trained visual embeddings as a semantically and stylistically aware feature space, measure cosine similarity, and validate outputs through inspection tools.

2. Pipeline Design

The system is implemented as modular Python scripts (src/) with an inspection notebook (notebooks/demo_pipeline.ipynb).

Data ingestion and frame extraction:

Videos placed in data/raw/ are sampled at fixed intervals (e.g., every 2 seconds). Frames are stored in data/frames/<video_id>/, with metadata logged in outputs/frames_index.csv.

Embedding computation:

Frames are processed using the HuggingFace CLIP model (openai/clip-vit-base-patch32). Fixed-size embeddings (e.g., 512-d) are L2-normalized and stored in outputs/embeddings.npy, with traceable metadata in outputs/embeddings_index.csv.

Similarity and retrieval:

Pairwise cosine similarity is computed, generating:

- outputs/topk.csv (nearest neighbors)
- outputs/heatmap.png (similarity matrix)
- outputs/tsne.png (2D embedding projection)

Design choices prioritize reproducibility, simplicity, and traceability. L2 normalization ensures cosine similarity equals dot product. Metadata indexing guarantees all embeddings can be traced back to original frames and timestamps.

3. Verification and Testing

Verification was embedded at every stage.

Embedding checks:

- Shape validation: (N_frames, D)
- NaN/Inf assertions
- float32 dtype enforcement
- L2 norm ≈ 1 validation
- Identical-image embedding equality test

Similarity invariants:

- Symmetry ($S \approx S.T$)
- Diagonal ≈ 1
- Values constrained to $[-1, 1]$

Unit tests verify frame counts, embedding dimensionality, and numerical integrity. Manual inspection of heatmaps and top-k neighbor grids ensures perceptual alignment with stylistic expectations.

4. AI Usage and Governance

GitHub Copilot was used for repetitive boilerplate code (CLI scaffolding, plotting templates). ChatGPT was used for conceptual validation and documentation refinement.

All AI-generated code was manually reviewed and wrapped with assertions or tests where applicable. An `ai_assistant_log.md` documents representative prompts, edits, and justification. Architectural decisions and validation strategies were not delegated to AI systems.

5. Sample Results

Similarity heatmaps reveal high-similarity blocks corresponding to stylistically coherent scenes.

Top-k retrieval frequently surfaces frames with similar color palette, texture, or composition, suggesting embeddings capture stylistic “vibe” beyond object identity.

t-SNE projections show clustering by mood or dominant visual characteristics. A fixed random seed ensures reproducibility.

6. Next Steps

To evolve into a production GACS engine:

1. **Multimodal fusion:** integrate audio and text/OCR embeddings.
2. **Human calibration:** collect vibe annotations for supervised fine-tuning.
3. **Performance linkage:** model embeddings against CTR/ROAS.
4. **Productionization:** store embeddings in a vector database (e.g., FAISS/Milvus) with API access.
5. **Online evaluation:** refine embedding-performance mappings through A/B testing.

7. Reproducibility

Environment: Python 3.9+, Conda recommended.

Typical workflow:

- Extract frames
- Compute embeddings
- Compute similarity and generate visualizations

All scripts include fail-fast assertions to prevent invalid execution.

Conclusion

This Mini GACS prototype demonstrates a compact, verifiable method for estimating stylistic similarity across art and marketing video frames. It combines pre-trained visual embeddings with robust similarity logic, structured validation, and traceable metadata.

The modular design and defined roadmap provide a clear path toward scaling into a multimodal, production-grade GACS system capable of linking creative “vibe” to measurable performance metrics.

Repository: <https://github.com/Nabanshu-Barman/genTA-mini-gacs>

End of Report